

Superstore Profitability Analysis

Tools: SQL, Power BI

Dataset: Sample Superstore

Goal: Identify the drivers of low profitability and propose focus areas for management.

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Chapter 1 - Business Overview

Business Objective

The objective of this analysis is to evaluate the overall business performance, identify the strongest and weakest areas of the business, and determine which aspects require deeper investigation.

Query 1.1 – Overall KPIs

1 SELECT

2 SUM(sales) AS total_sales,

3 SUM(profit) AS total_profit,

4 ROUND(SUM(profit) / SUM(sales) *100,2) AS profit_margin,

5 SUM(quantity) AS total_quantity,

6 COUNT(DISTINCT customer_id) AS total_customers,

7 COUNT(DISTINCT order_id) AS total_orders

8 FROM superstore

Data Output

Messages

Notifications

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SQL

	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	2296957.7003	286324.0737	12.47	37871	793	5008

Query 1.2 – Category Performance

category text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
Furniture	741999.7953	18451.2728	2.49	8028	707	1764
Office Supplies	718803.872	122417.8528	17.03	22904	788	3741
Technology	836154.033	145454.9481	17.40	6939	687	1544

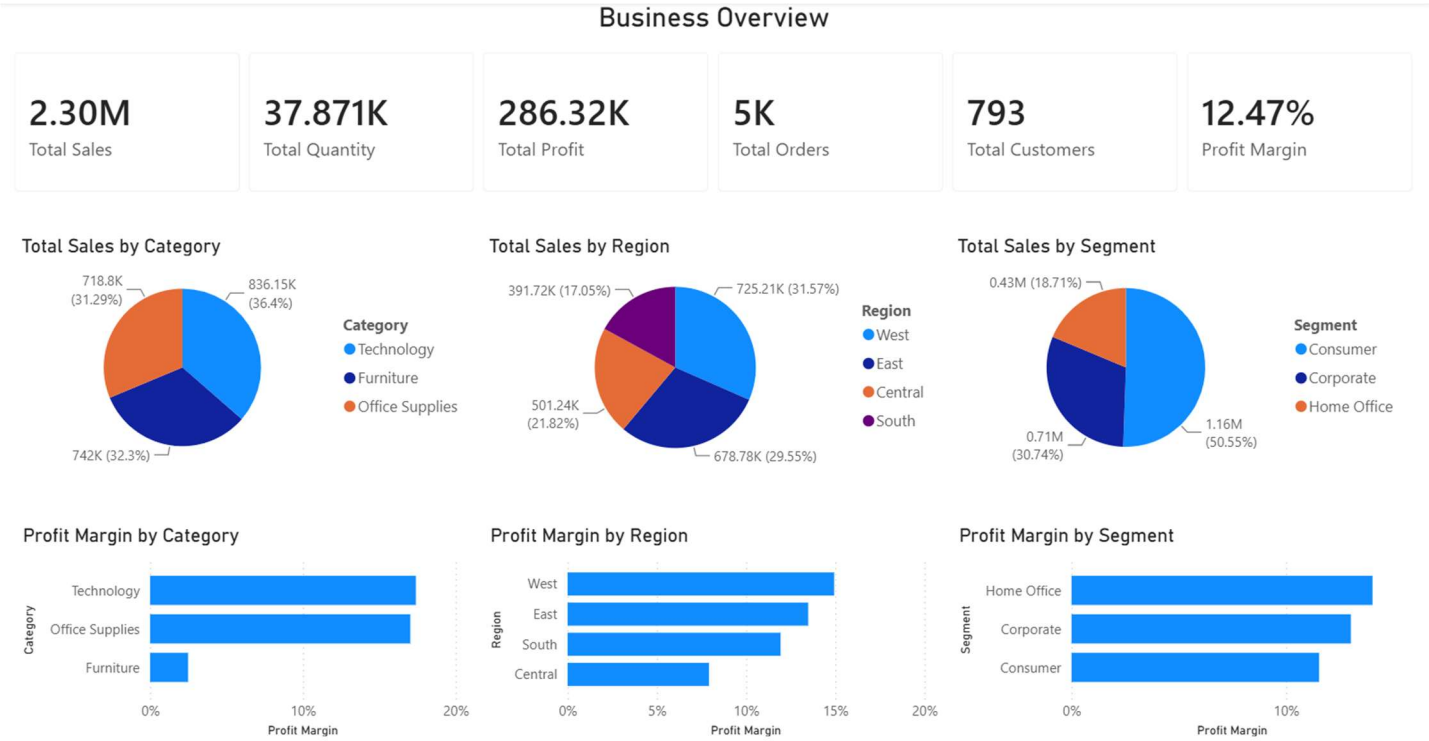
Query 1.3 – Segment Performance

	segment text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	Consumer	1161158.1850	134046.2612	11.54	19519	409	2585
2	Corporate	706146.3668	91979.1340	13.03	11608	236	1514
3	Home Office	429653.1485	60298.6785	14.03	6744	148	909

Query 1.4 – Regional Performance

	region text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	Central	501239.8908	39706.3625	7.92	8780	629	1175
2	East	678781.240	91522.7800	13.48	10618	674	1401
3	South	391721.9050	46749.4303	11.93	6209	512	822
4	West	725214.6645	108345.5009	14.94	12264	686	1610

Dashboard 1.1 – Business Overview



Chapter 2 – Category & Product Performance Analysis

Business Objective

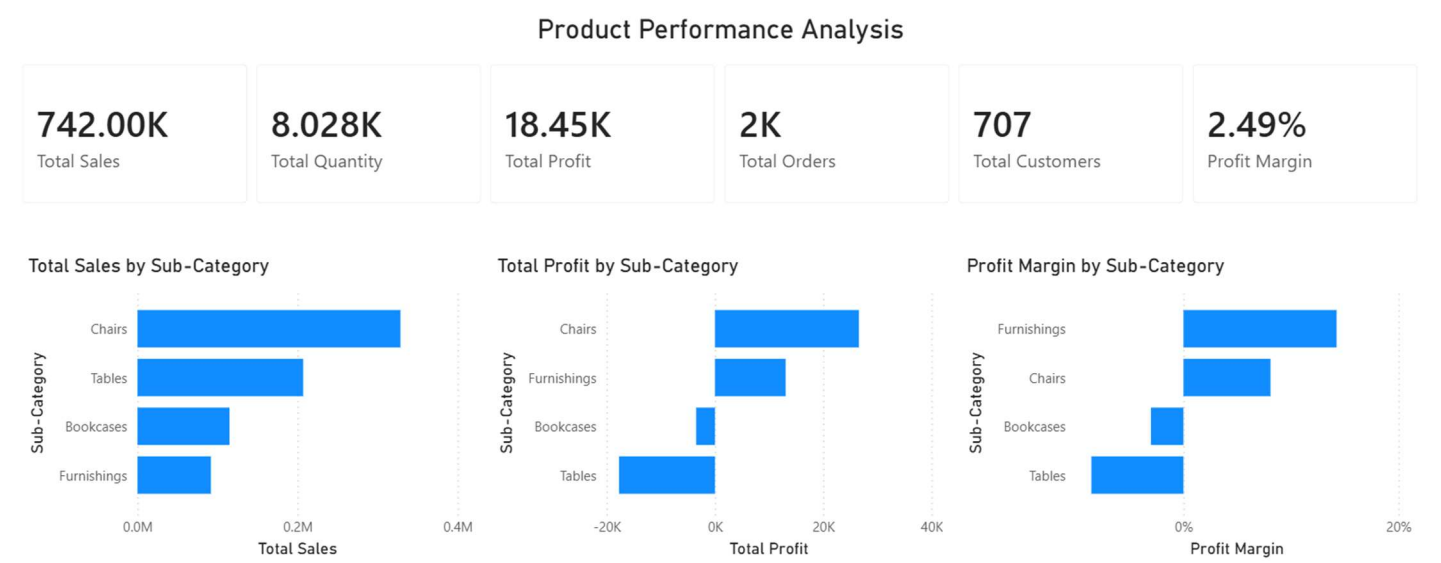
Analyze category and sub-category performance to identify profitability risks and priorities for further analysis.

Query 2.1 – Category & Product Performance

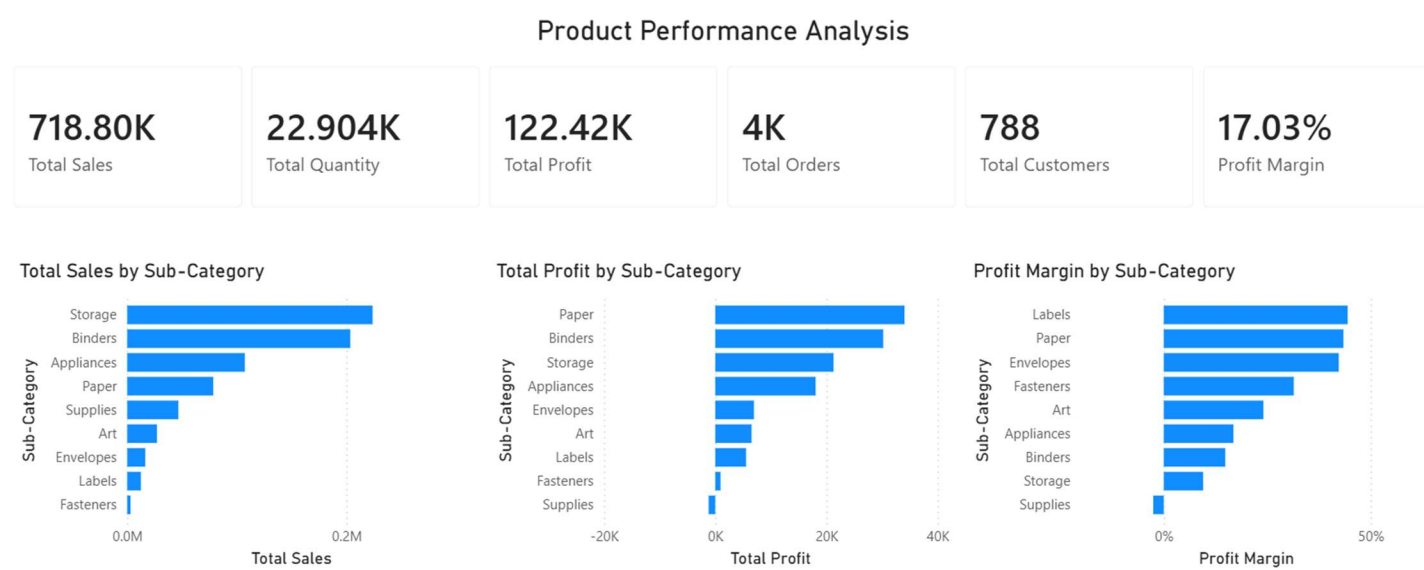
```
1 SELECT category, sub_category,
2 SUM (sales) AS total_sales,
3 SUM (profit) AS total_profit,
4 ROUND (SUM (profit) / SUM (sales) * 100,2) AS profit_margin,
5 SUM (quantity) AS total_quantity,
6 COUNT (DISTINCT customer_id) AS total_customers,
7 COUNT (DISTINCT order_id) AS total_orders
8 FROM superstore
9 GROUP BY category, sub_category
10 ORDER BY category, profit_margin DESC
```

	category text	sub_category text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	Furniture	Furnishings	91705.164	13059.1436	14.24	3563	528	877
2	Furniture	Chairs	328449.103	26590.1663	8.10	2356	407	576
3	Furniture	Bookcases	114879.9963	-3472.5560	-3.02	868	195	224
4	Furniture	Tables	206965.5320	-17725.4811	-8.56	1241	261	307
5	Office Supplies	Labels	12486.312	5546.2540	44.42	1400	281	346
6	Office Supplies	Paper	78479.206	34053.5693	43.39	5178	611	1191
7	Office Supplies	Envelopes	16476.402	6964.1767	42.27	906	206	249
8	Office Supplies	Fasteners	3024.280	949.5182	31.40	914	191	215
9	Office Supplies	Art	27118.792	6527.7870	24.07	3000	494	731
10	Office Supplies	Appliances	107289.001	18065.0574	16.84	1727	356	450
11	Office Supplies	Binders	203412.733	30221.7633	14.86	5974	650	1316
12	Office Supplies	Storage	223843.608	21278.8264	9.51	3158	514	777
13	Office Supplies	Supplies	46673.538	-1189.0995	-2.55	647	160	187
14	Technology	Copiers	149528.030	55617.8249	37.20	234	64	68
15	Technology	Accessories	167380.318	41936.6357	25.05	2976	474	718
16	Technology	Phones	330007.054	44515.7306	13.49	3289	511	814
17	Technology	Machines	189238.631	3384.7569	1.79	440	99	112

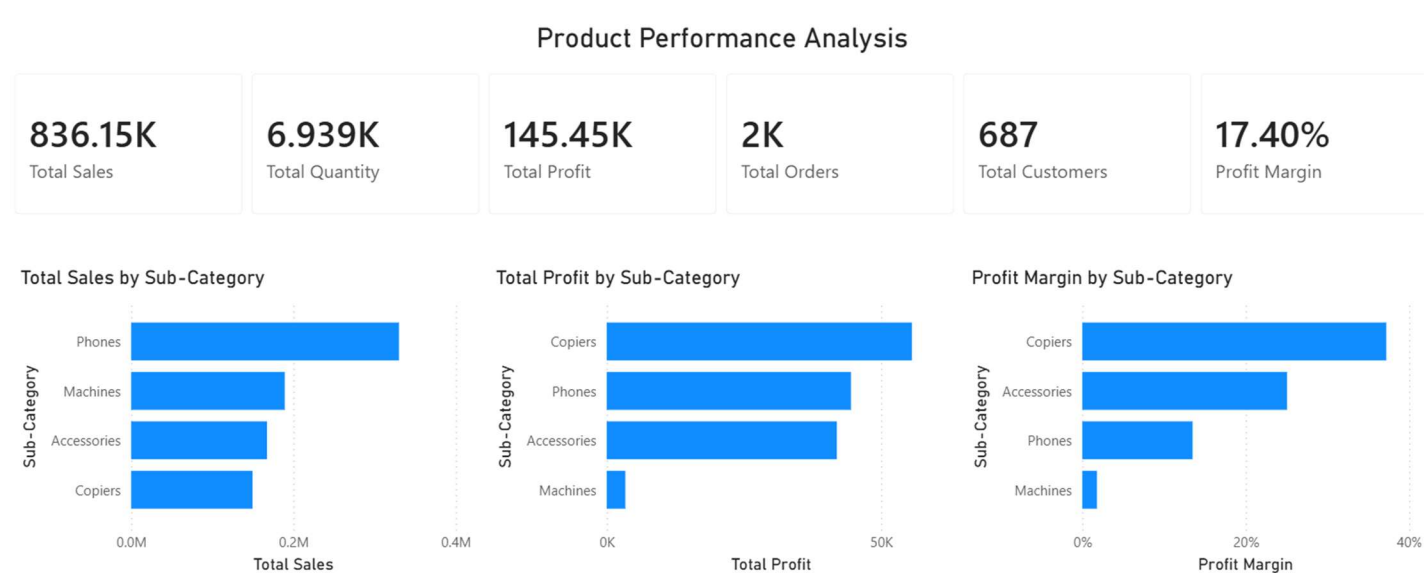
Dashboard 2.1 – Furniture Performance Analysis



Dashboard 2.2 - Office Supplies Performance Analysis



Dashboard 2.3 - Technology Performance Analysis



Key Insights: Furniture is the weakest category, driven by loss-making Tables and Bookcases. These sub-categories were selected for further discount analysis.

Which Furniture sub-categories contribute most to low profitability?

High-Revenue Product Profitability

The purpose of this comparison is to determine whether the highest-selling products also generate healthy profits across each category.

Query 2.2 - Technology products top 10 by sales:

	product_id text	sub_category text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	TEC-CO-10004722	Copiers	61599.824	25199.9280	40.91	20	5	5
2	TEC-MA-10002412	Machines	22638.48	-1811.0784	-8.00	6	1	1
3	TEC-CO-10001449	Copiers	18839.686	6983.8836	37.07	38	8	8
4	TEC-MA-10001127	Machines	18374.895	4094.9766	22.29	12	3	3
5	TEC-MA-10000822	Machines	16829.901	-4589.973	-27.27	18	4	4
6	TEC-MA-10001047	Machines	14299.89	3717.9714	26.00	11	2	2
7	TEC-PH-10001459	Phones	13943.668	1822.7566	13.07	39	6	6
8	TEC-AC-10002049	Accessories	13756.536	4425.3432	32.17	60	14	15
9	TEC-PH-10002885	Phones	12996.600	1611.5784	12.40	24	6	6
10	TEC-CO-10003763	Copiers	11619.834	4570.9347	39.34	19	4	4

Query 2.3 - Office Supplies products top 10 by sales:

	product_id text	sub_category text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	OFF-BI-10003527	Binders	27453.384	7753.039	28.24	31	10	10
2	OFF-BI-10001359	Binders	19823.479	2233.5051	11.27	37	11	11
3	OFF-BI-10000545	Binders	19024.500	760.980	4.00	48	9	9
4	OFF-BI-10004995	Binders	17965.068	-1878.1662	-10.45	27	6	6
5	OFF-SU-10000151	Supplies	17030.312	-262.0048	-1.54	11	3	3
6	OFF-SU-10002881	Supplies	16656.200	-1299.1836	-7.80	22	6	6
7	OFF-BI-10001120	Binders	15875.916	3345.2823	21.07	13	3	3
8	OFF-AP-10002945	Appliances	11304.44	3247.0200	28.72	40	8	8
9	OFF-ST-10000078	Storage	10925.004	1575.1098	14.42	43	11	11
10	OFF-ST-10003716	Storage	9630.856	429.7882	4.46	47	12	12

Query 2.4 - Furniture products top 10 by sales:

	product_id text	sub_category text	total_sales numeric	total_profit numeric	profit_margin numeric	total_quantity bigint	total_customers bigint	total_orders bigint
1	FUR-CH-10002024	Chairs	21870.576	0.000	0.00	39	8	8
2	FUR-BO-10004834	Bookcases	15610.9656	-669.5448	-4.29	24	5	5
3	FUR-TA-10003473	Tables	12995.2915	-327.2331	-2.52	46	11	12
4	FUR-CH-10001215	Chairs	12975.382	951.862	7.34	32	9	9
5	FUR-BO-10002213	Bookcases	12921.643	19.6864	0.15	42	10	10
6	FUR-CH-10004287	Chairs	11572.78	1179.374	10.19	53	13	13
7	FUR-CH-10000454	Chairs	10637.528	1927.442	18.12	51	12	12
8	FUR-TA-10000198	Tables	9917.640	-2876.1156	-29.00	27	5	5
9	FUR-TA-10001889	Tables	9544.725	-1934.3976	-20.27	33	7	7
10	FUR-CH-10003973	Chairs	9070.944	773.2608	8.52	31	8	8

Key Observations

- Technology includes several high-revenue and highly profitable products, although some expensive Machines generate losses.
- Office Supplies demonstrates relatively balanced profitability among top-selling products.
- Furniture contains multiple high-selling products with negative profit margins, confirming that strong sales do not necessarily translate into profitability.

Business Question for Further Investigation: Why are Tables and Bookcases unprofitable, and how does discount level relate to their profitability?

Chapter 3 – Discount Analysis

Business Objective

Analyze how discount levels affect profitability overall, by category, and by sub-category.

Query 3.1 – Overall Discount Performance

Finding: Overall profitability turns negative at a 30% discount, suggesting that discounts at or above this level require closer control.

discount numeric	sum_sales numeric	sum_profit numeric	profit_margin numeric
0	1087665.31	320914.6552	29.50
0.1	54369.351	9029.1770	16.61
0.15	27558.5215	1418.9915	5.15
0.2	764594.368	90337.3060	11.82
0.3	103226.655	-10369.2774	-10.05
0.32	14493.4588	-2391.1377	-16.50
0.4	116417.784	-23057.0504	-19.81
0.45	5484.9740	-2493.1111	-45.45
0.5	58918.540	-20506.4281	-34.80
0.6	6644.700	-5944.6552	-89.46
0.7	40620.282	-40075.3569	-98.66
0.8	16963.756	-30539.0392	-180.03

Query 3.2 – Category Performance by Discount

Finding: Furniture becomes unprofitable at a lower discount threshold than the other categories.

category text	discount numeric	sum_sales numeric	sum_profit numeric	profit_margin numeric
Furniture	0	256025.27	58133.0764	22.71
Furniture	0.1	46634.247	7111.0119	15.25
Furniture	0.15	27558.5215	1418.9915	5.15
Furniture	0.2	216631.016	6265.9491	2.89
Furniture	0.3	99470.350	-10695.3169	-10.75
Furniture	0.32	14493.4588	-2391.1377	-16.50
Furniture	0.4	45614.406	-16187.3968	-35.49
Furniture	0.45	5484.9740	-2493.1111	-45.45
Furniture	0.5	20983.470	-12871.1990	-61.34
Furniture	0.6	6644.700	-5944.6552	-89.46
Furniture	0.7	2459.382	-3894.9394	-158.37
Office Supplies	0	441906.84	130433.1584	29.52
Office Supplies	0.1	4324.149	1086.0808	25.12
Office Supplies	0.2	233049.736	38038.7512	16.32
Office Supplies	0.7	22559.391	-16601.0984	-73.59
Office Supplies	0.8	16963.756	-30539.0392	-180.03
Technology	0	389733.20	132348.4204	33.96
Technology	0.1	3410.955	832.0843	24.39
Technology	0.2	314913.616	46032.6057	14.62
Technology	0.3	3756.305	326.0395	8.68
Technology	0.4	70803.378	-6869.6536	-9.70
Technology	0.5	37935.070	-7635.2291	-20.13
Technology	0.7	15601.509	-19579.3191	-125.50

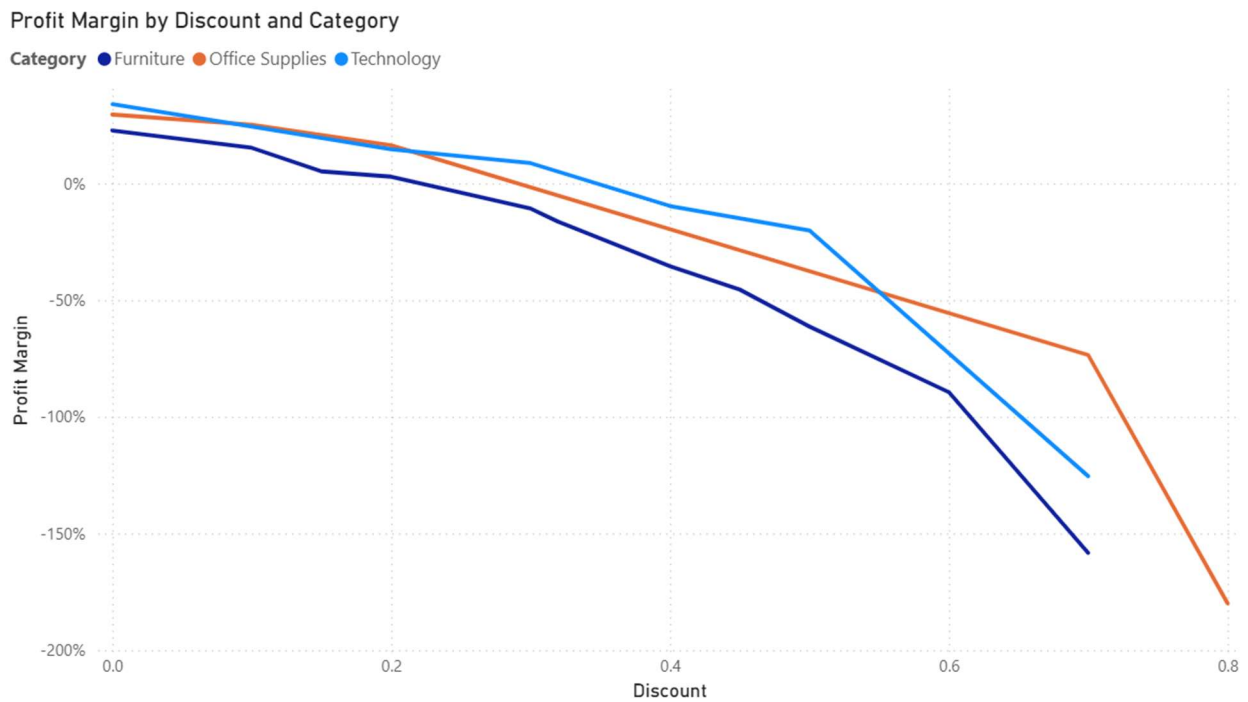
Query 3.3 – Sub-Category Performance by Discount

category text	sub_category text	discount numeric	sum_sales numeric	sum_profit numeric	profit_margin numeric
Furniture	Bookcases	0	31935.98	6075.7117	19.02
Furniture	Bookcases	0.15	27558.5215	1418.9915	5.15
Furniture	Bookcases	0.2	26841.496	130.5022	0.49
Furniture	Bookcases	0.3	4282.698	-555.8726	-12.98
Furniture	Bookcases	0.32	14493.4588	-2391.1377	-16.50
Furniture	Bookcases	0.5	7308.460	-4255.8117	-58.23
Furniture	Bookcases	0.7	2459.382	-3894.9394	-158.37
Furniture	Chairs	0	91060.73	21933.0961	24.09
Furniture	Chairs	0.1	46634.247	7111.0119	15.25
Furniture	Chairs	0.2	120748.624	4283.1750	3.55
Furniture	Chairs	0.3	70005.502	-6737.1167	-9.62
Furniture	Furnishings	0	61449.80	16847.9689	27.42
Furniture	Furnishings	0.2	23610.664	2155.8299	9.13
Furniture	Furnishings	0.6	6644.700	-5944.6552	-89.46
Furniture	Tables	0	71578.76	13276.2997	18.55
Furniture	Tables	0.2	45430.232	-303.5580	-0.67
Furniture	Tables	0.3	25182.150	-3402.3276	-13.51
Furniture	Tables	0.4	45614.406	-16187.3968	-35.49
Furniture	Tables	0.45	5484.9740	-2493.1111	-45.45
Furniture	Tables	0.5	13675.010	-8615.3873	-63.00

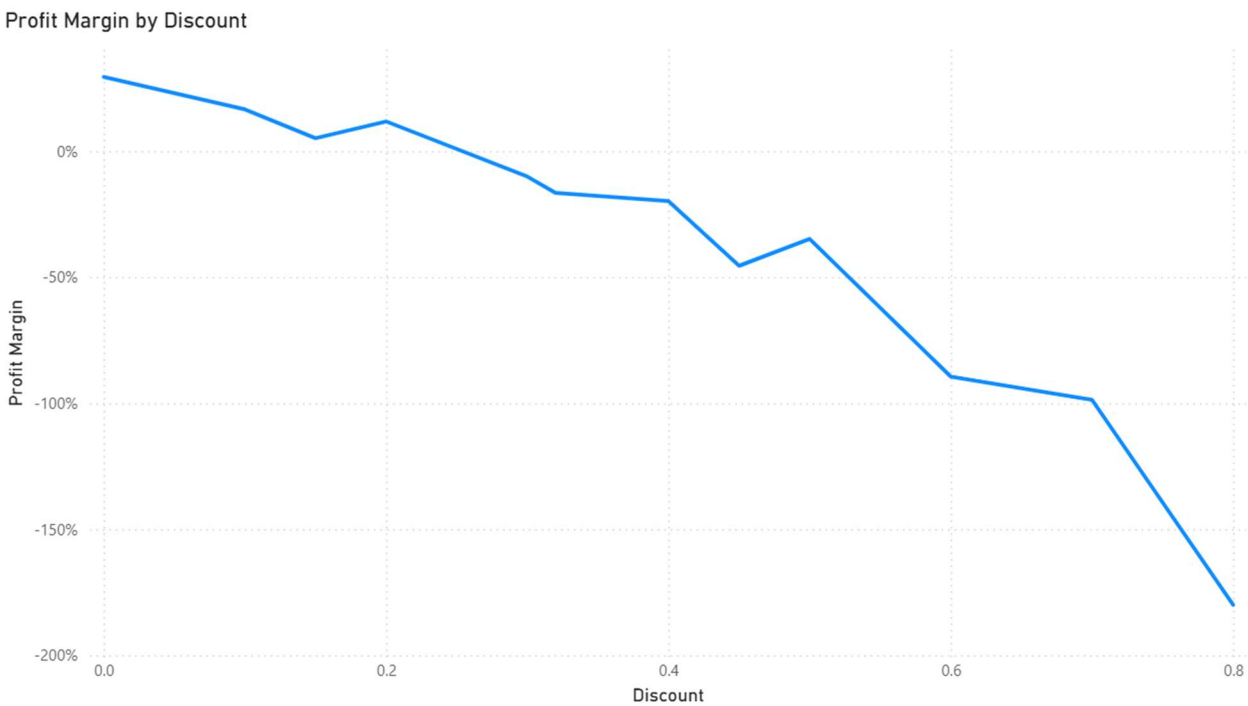
category text	sub_category text	discount numeric	sum_sales numeric	sum_profit numeric	profit_margin numeric
Office Supplies	Appliances	0	77823.03	23110.7881	29.70
Office Supplies	Appliances	0.1	4324.149	1086.0808	25.12
Office Supplies	Appliances	0.2	21759.288	2497.8297	11.48
Office Supplies	Appliances	0.8	3382.534	-8629.6412	-255.12
Office Supplies	Art	0	18014.64	5380.6006	29.87
Office Supplies	Art	0.2	9104.152	1147.1864	12.60
Office Supplies	Binders	0	81829.48	39314.4507	48.04
Office Supplies	Binders	0.2	85442.640	29417.8090	34.43
Office Supplies	Binders	0.7	22559.391	-16601.0984	-73.59
Office Supplies	Binders	0.8	13581.222	-21909.3980	-161.32
Office Supplies	Envelopes	0	10606.45	4976.9844	46.92
Office Supplies	Envelopes	0.2	5869.952	1987.1923	33.85
Office Supplies	Fasteners	0	1822.84	652.2052	35.78
Office Supplies	Fasteners	0.2	1201.440	297.3130	24.75
Office Supplies	Labels	0	9239.56	4422.0960	47.86
Office Supplies	Labels	0.2	3246.752	1124.1580	34.62
Office Supplies	Paper	0	53157.87	25329.4677	47.65
Office Supplies	Paper	0.2	25321.336	8724.1016	34.45
Office Supplies	Storage	0	157853.76	25528.1715	16.17
Office Supplies	Storage	0.2	65989.848	-4249.3451	-6.44
Office Supplies	Supplies	0	31559.21	1718.3942	5.44
Office Supplies	Supplies	0.2	15114.328	-2907.4937	-19.24

category text	sub_category text	discount numeric	sum_sales numeric	sum_profit numeric	profit_margin numeric
Technology	Accessories	0	118370.31	35289.2539	29.81
Technology	Accessories	0.2	49010.008	6647.3818	13.56
Technology	Copiers	0	76449.18	35556.1349	46.51
Technology	Copiers	0.2	56159.048	17878.7148	31.84
Technology	Copiers	0.4	16919.802	2182.9752	12.90
Technology	Machines	0	71034.00	27137.8223	38.20
Technology	Machines	0.1	3410.955	832.0843	24.39
Technology	Machines	0.2	37954.568	4970.2024	13.10
Technology	Machines	0.3	3756.305	326.0395	8.68
Technology	Machines	0.4	19546.224	-2666.8434	-13.64
Technology	Machines	0.5	37935.070	-7635.2291	-20.13
Technology	Machines	0.7	15601.509	-19579.3191	-125.50
Technology	Phones	0	123879.71	34365.2093	27.74
Technology	Phones	0.2	171789.992	16536.3067	9.63
Technology	Phones	0.4	34337.352	-6385.7854	-18.60

Visual 3.1 – Profit Margin by Discount and Category



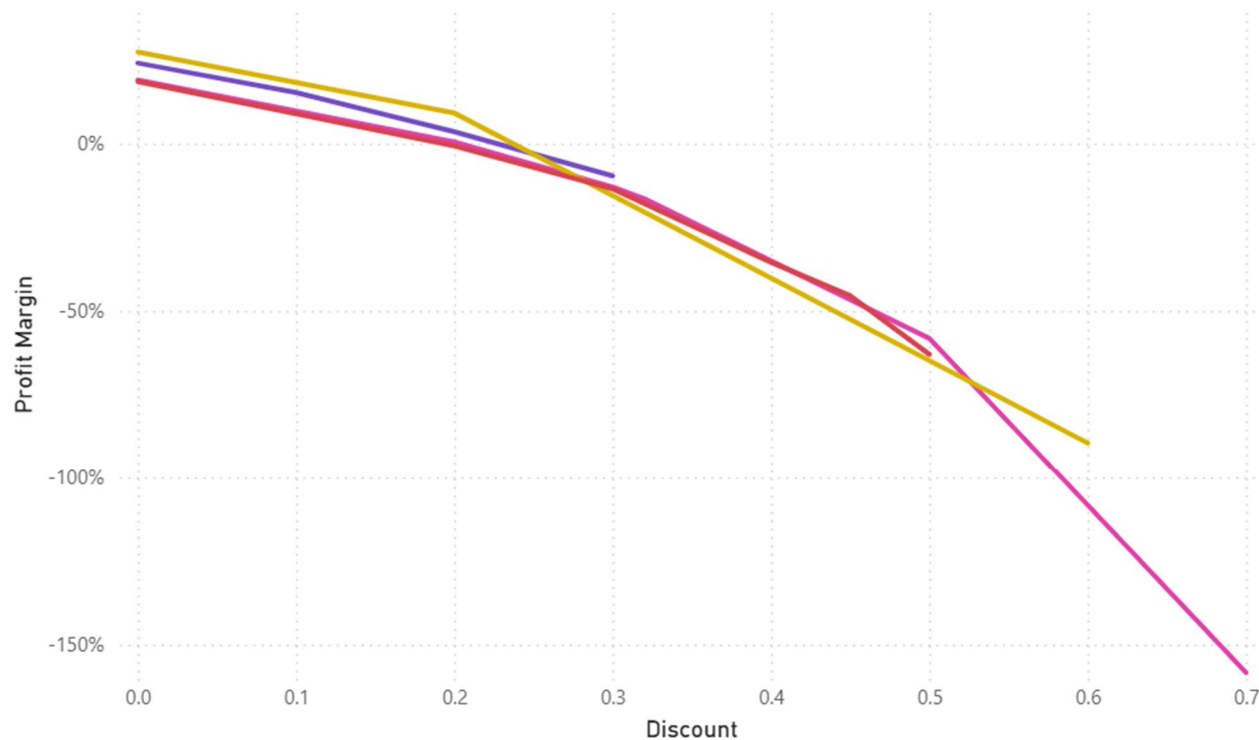
Visual 3.2 – Profit Margin by Discount



Visual 3.3 – Profit Margin by Discount and Sub-category

Profit Margin by Discount and Sub-Category

Sub-Category ● Bookcases ● Chairs ● Furnishings ● Tables



Key Insights

- In the observed data, sales at discount levels of 30% or higher have a negative aggregate profit margin.
- Furniture becomes unprofitable at a lower discount threshold than the other categories.
- Tables are the most discount-sensitive Furniture sub-category, becoming unprofitable at a 20% discount.

Observation

The impact of discounts is not uniform across the business. Losses are concentrated in specific sub-categories, particularly Tables, rather than occurring evenly across all products.

Business Question for Further Investigation

Does the Central region have a higher exposure to discount-sensitive sub-categories or higher discount levels, contributing to its low profit margin?

Chapter 4 – Region Analysis

Business Objective

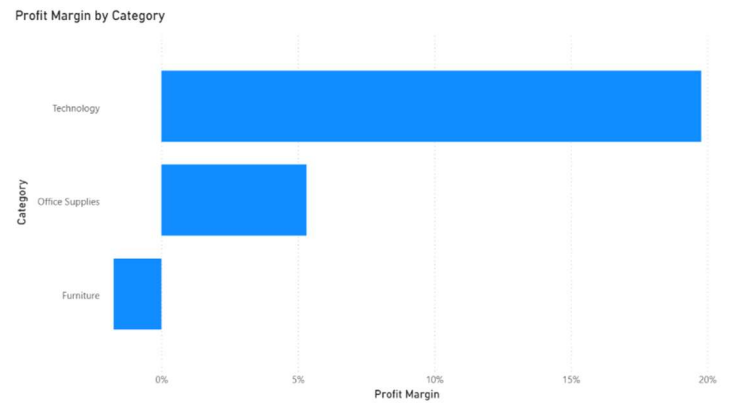
Analyze Central region performance to identify the categories, sub-categories, and discount patterns driving low profitability.

Query 4.1 - Category Performance in Central

Finding: Furniture is the only loss-making category in the Central region, with a profit margin of –1.75%.

category	sum_sales	sum_profit	profit_margin
text	numeric	numeric	numeric
Furniture	163797.1638	-2871.0494	-1.75
Office Supplies	167026.415	8879.9799	5.32
Technology	170416.312	33697.4320	19.77

Visual 4.1 - Profit Margin by Category — Central Region

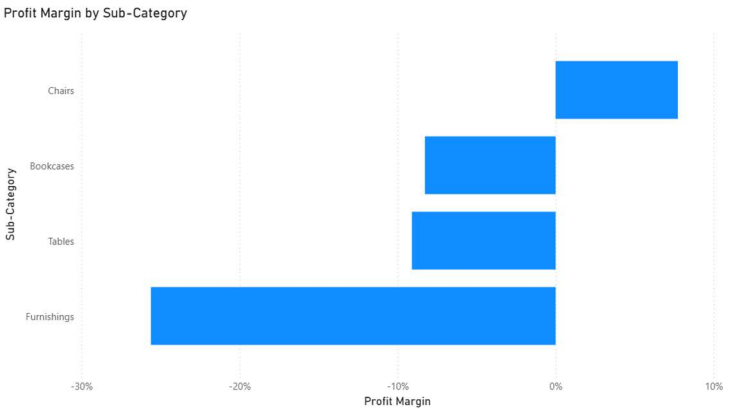


Query 4.2 - Furniture Sub-Category Performance in Central

Finding: Furnishings is the most unprofitable Furniture sub-category in Central, with a profit margin of –25.61%. Tables and Bookcases are also loss-making, while Chairs remains profitable.

	sub_category	sum_sales	sum_profit	profit_margin
	text	numeric	numeric	numeric
1	Furnishings	15254.370	-3906.2168	-25.61
2	Tables	39154.971	-3559.6504	-9.09
3	Bookcases	24157.1768	-1997.9043	-8.27
4	Chairs	85230.646	6592.7221	7.74

Visual 4.2 - Profit Margin by Sub-Category — Central Region



Query 4.3 - Furniture Sub-Category Performance by Discount in Central

Finding: Within Central Furniture, observed discounted sales are loss-making across all sub-categories.

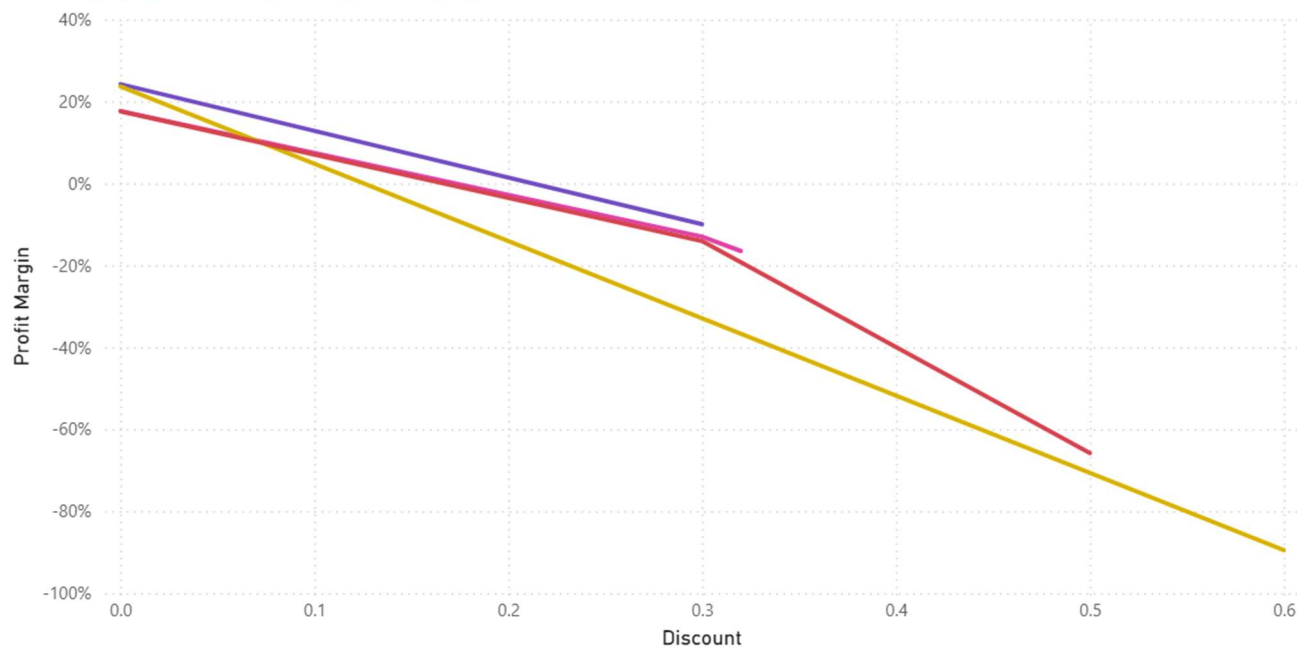
```
SELECT sub_category, discount,
SUM(sales) AS sum_sales,
SUM(profit) AS sum_profit,
ROUND(SUM(profit) / NULLIF(SUM(sales), 0)*100,2) AS profit_margin
FROM superstore
WHERE region = 'Central' AND category = 'Furniture'
GROUP BY sub_category, discount
ORDER BY profit_margin;
```

sub_category text	discount numeric	sum_sales numeric	sum_profit numeric	profit_margin numeric
Furnishings	0.6	6644.700	-5944.6552	-89.46
Tables	0.5	6550.670	-4309.7447	-65.79
Bookcases	0.32	14493.4588	-2391.1377	-16.50
Tables	0.3	15760.661	-2216.6766	-14.06
Bookcases	0.3	4282.698	-555.8726	-12.98
Chairs	0.3	41135.626	-4094.3445	-9.95
Tables	0	16843.64	2966.7709	17.61
Bookcases	0	5381.02	949.1060	17.64
Furnishings	0	8609.67	2038.4384	23.68
Chairs	0	44095.02	10687.0666	24.24

Visual 4.3 - Profit Margin by Discount and Furniture Sub-Category — Central Region

Profit Margin by Discount and Sub-Category

Sub-Category ● Bookcases ● Chairs ● Furnishings ● Tables



Observation: Central's low profitability is concentrated in Furniture and is associated with loss-making discounted sales.

Chapter 5 – Executive Summary & Recommendations

Executive Summary

Superstore is profitable overall, with a profit margin of 12.47%. However, profitability is weakened by Furniture, which generates sales comparable to Technology but has a substantially lower margin.

The analysis identified two connected risk areas: discount-sensitive Furniture sub-categories, particularly Tables, and the Central region, where Furniture is the only loss-making category. Within Central, Furnishings, Tables, and Bookcases all generate negative profit margins.

Recommended Actions

1. Review discount policy for Furniture, with priority given to Tables, which show a negative aggregate profit margin at a 20% discount in the observed data.
2. Investigate pricing and cost structure for loss-making Furniture in Central, especially Furnishings, Tables, and Bookcases.
3. Monitor profit margin by region, category, sub-category, and discount level rather than relying on sales volume alone.

Note: This analysis identifies relationships in historical sales data. Further validation would require product-level cost and order-volume data.